

Vectors, distances, and dissimilarity matrices

The linear algebra behind the first lectures and Assignment 1, with the numbers written out

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Who this is for

Several of you said that the linear algebra in the first lectures went by quickly. This handout takes each object one at a time, defines it, and works a small numerical example by hand. Start with the worked examples; after each one, cover the calculation and try it yourself. Vectors here have two or three entries so the arithmetic fits on a line. The same formulas apply when the vectors have thousands of entries.

Similarity as distance. A list of D numbers is a point in a D -dimensional space. We can model the similarity of two things by choosing a representation, so that each thing is a point, and a rule for measuring dissimilarity between points. There is more than one reasonable rule; this handout works through three.

How to read the symbols. $\sum_{j=1}^D$ means “repeat the expression that follows for $j = 1, 2, \dots, D$, then add the results”. $x \in [0, 2]$ means x lies between 0 and 2, endpoints included; $[0, \infty)$ means $x \geq 0$. $|c|$ is the absolute value of c . $\arccos$ undoes the cosine: $\arccos(0.96)$ is the angle whose cosine is 0.96. Vector entries are written horizontally to save space; in a matrix–vector product (§9) the input vector is a column, and in a data matrix each response vector occupies one row.

1 Vectors, and the names the course uses

1.1 A vector is a list of numbers, and also a point

A *vector* is an ordered list of numbers. Write it in bold, $\mathbf{x}$, and write its j -th entry as x_j :

$$\mathbf{x} = (x_1, x_2, \dots, x_D).$$

D is the number of entries, the *dimension*. Two readings of the same list are worth holding at once. As a **point**, $\mathbf{x}$ is a location: with $D = 2$, $(3, 4)$ is the point three units right and four units up. As an **arrow**, $\mathbf{x}$ is the displacement from the origin to that point, with a length and a direction. The lectures use both.

Example. Record the firing rates of three neurons to one image and get $\mathbf{x} = (2, 0, 5)$: neuron 1 fired at 2 spikes per second, neuron 2 was silent, neuron 3 fired at 5. That image is now a point in a three-dimensional “neuron space”, one axis per neuron. Pixel values, a row of network activations, and a set of ratings are all vectors in exactly this sense.

1.2 Stimuli are indexed with a superscript in parentheses

The course looks at many stimuli at once, so it needs two indices: *which stimulus* and *which entry*. The convention in the lectures:

- $\mathbf{x}^{(i)}$ is the response vector to stimulus i , for $i = 1, \dots, N$;
- $x_j^{(i)}$ is its j -th entry, for $j = 1, \dots, D$.

Here N is the number of represented items (images, conditions) and D is the number of features (pixels, units, neurons, rating scales). The Assignment 1 notebooks call these dimensions n and p . The parentheses in $\mathbf{x}^{(i)}$ identify an item; they do not indicate a power. Read $x_3^{(7)}$ as “the response of neuron 3 to image 7”.

1.3 Stacking the vectors: the response matrix

Put the N response vectors on top of each other, one per row, and you get the *response matrix* $\mathbf{X}$, with N rows and D columns:

$$\mathbf{X} = \begin{pmatrix} -\mathbf{x}^{(1)} - \\ -\mathbf{x}^{(2)} - \\ \vdots \\ -\mathbf{x}^{(N)} - \end{pmatrix} \quad \mathbf{X} \text{ is } N \times D, \quad (\mathbf{X})_{ij} = x_j^{(i)}.$$

Example. Three images, two neurons ($N = 3$, $D = 2$), with $\mathbf{x}^{(1)} = (3, 4)$, $\mathbf{x}^{(2)} = (4, 3)$, $\mathbf{x}^{(3)} = (0, 1)$:

$$\mathbf{X} = \begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 0 & 1 \end{pmatrix}, \quad \text{row 2 is } \mathbf{x}^{(2)}, \quad \text{column 2 is neuron 2's response to every image.}$$

“ $N \times D$ ” always means rows first, then columns. For the response matrices in this course, *rows are stimuli and columns are features*.

Mathematics counts from 1 and Python counts from 0: the mathematical $\mathbf{x}^{(1)}$ is $\mathbf{X}[0]$, and $x_j^{(i)}$ is $\mathbf{X}[i-1, j-1]$.

In NumPy: $\mathbf{X}.\text{shape}$ is (N, D) ; $\mathbf{X}[0]$ is the first row; $\mathbf{X}[:, 0]$ is the first column.

Try it. Which entry of $\mathbf{X}$ above is neuron 2's response to image 3? (Answers at the end.)

2 Adding, subtracting, and scaling vectors

Vectors of the same dimension are added and subtracted entry by entry, and multiplied by a number (a *scalar*) entry by entry:

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_D + y_D), \quad \mathbf{x} - \mathbf{y} = (x_1 - y_1, \dots, x_D - y_D), \quad c\mathbf{x} = (cx_1, \dots, cx_D).$$

Example. With $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (4, 3)$:

$$\mathbf{x} + \mathbf{y} = (7, 7), \quad \mathbf{x} - \mathbf{y} = (-1, 1), \quad 2\mathbf{x} = (6, 8), \quad \frac{1}{2}\mathbf{x} = (1.5, 2).$$

Geometrically, $\mathbf{x} + \mathbf{y}$ is the arrow you get by placing $\mathbf{y}$'s tail at $\mathbf{x}$'s head. For $c > 0$, $c\mathbf{x}$ points the same way as $\mathbf{x}$ and is c times as long; for $c < 0$ it points the opposite way and is $|c|$ times as long; for $c = 0$ it is the zero vector $\mathbf{0} = (0, \dots, 0)$, which has no direction.

The difference vector. $\mathbf{x} - \mathbf{y}$ is the arrow from the point $\mathbf{y}$ to the point $\mathbf{x}$, and its length is the distance between them. For example, $(3, 4) - (4, 3) = (-1, 1)$: from $(4, 3)$, move one unit left and one unit up to reach $(3, 4)$. Subtracting two response vectors gives the displacement between two stimuli in neuron space; §4 measures its length.

In NumPy: `x + y`, `x - y`, `2 * x`, all entry by entry.

3 The dot product

3.1 Definition

The *dot product* of two vectors of the same dimension multiplies matching entries and adds the results. It turns two vectors into one number:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^D x_j y_j = x_1 y_1 + x_2 y_2 + \dots + x_D y_D.$$

Example. $(3, 4) \cdot (4, 3) = 3 \cdot 4 + 4 \cdot 3 = 12 + 12 = 24$. $(3, 4) \cdot (0, 1) = 0 + 4 = 4$. $(3, 4) \cdot (4, -3) = 12 - 12 = 0$.

The dot product is symmetric, $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$, and a vector dotted with itself gives the sum of its squared entries: $(3, 4) \cdot (3, 4) = 9 + 16 = 25$.

3.2 The Euclidean norm, or length

The *norm* of $\mathbf{x}$ is its length as an arrow, by Pythagoras:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_j x_j^2}.$$

Example. $\|(3, 4)\| = \sqrt{9 + 16} = 5$. $\|(4, 3)\| = 5$. $\|(0, 1)\| = 1$. $\|(1, 1, 1)\| = \sqrt{3} \approx 1.732$.

Scaling a vector scales its length: $\|c\mathbf{x}\| = |c| \|\mathbf{x}\|$, so $\|2(3, 4)\| = \|(6, 8)\| = 10$. Dividing a vector by its own norm gives a *unit vector*, length exactly 1, pointing the same way: $(3, 4)/5 = (0.6, 0.8)$, and indeed $0.36 + 0.64 = 1$. This requires $\mathbf{x} \neq \mathbf{0}$: the zero vector has length zero and no direction, so it cannot be normalised.

In NumPy: `np.linalg.norm(x)`; `np.dot(x, y)` or `x @ y` for the dot product.

3.3 The geometric meaning: the angle between two vectors

The dot product is also determined by the two lengths and the angle ϕ between the arrows:

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \phi, \quad \text{equivalently} \quad \cos \phi = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

This is the fact that makes the dot product useful. For fixed vector lengths, the dot product is largest when the arrows point in the same direction ($\phi = 0$, $\cos \phi = 1$), zero when they are perpendicular ($\phi = 90^\circ$, $\cos \phi = 0$), and negative when the angle exceeds 90° . Its magnitude also depends on both lengths. The second formula needs both lengths to be nonzero.

Example. For $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (4, 3)$,

$$\cos \phi = \frac{24}{5 \cdot 5} = 0.96, \quad \phi = \arccos(0.96) \approx 16.3^\circ.$$

For $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (4, -3)$ the dot product was 0, so $\cos \phi = 0$ and the arrows are perpendicular; draw both on graph paper to confirm it. For $\mathbf{x} = (3, 4)$ and $2\mathbf{x} = (6, 8)$, $\cos \phi = 50/(5 \cdot 10) = 1$: same direction, angle zero.

4 Euclidean distance

The *Euclidean distance* between two points is the length of the difference vector, the straight-line distance between them:

$$d_{\text{euc}}(\mathbf{x}^{(i)}, \mathbf{x}^{(k)}) = \|\mathbf{x}^{(i)} - \mathbf{x}^{(k)}\| = \sqrt{\sum_j (x_j^{(i)} - x_j^{(k)})^2}.$$

Example. With $\mathbf{x}^{(1)} = (3, 4)$, $\mathbf{x}^{(2)} = (4, 3)$, $\mathbf{x}^{(3)} = (0, 1)$:

$$d_{\text{euc}}(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \|(-1, 1)\| = \sqrt{2} \approx 1.414,$$

$$d_{\text{euc}}(\mathbf{x}^{(1)}, \mathbf{x}^{(3)}) = \|(3, 3)\| = \sqrt{18} \approx 4.243,$$

$$d_{\text{euc}}(\mathbf{x}^{(2)}, \mathbf{x}^{(3)}) = \|(4, 2)\| = \sqrt{20} \approx 4.472.$$

So images 1 and 2 are close, and image 3 is far from both. This is the distance you would measure with a ruler on a drawing of the three points.

Three properties, all visible in the example: the distance from a point to itself is 0; $d_{\text{euc}}(\mathbf{x}, \mathbf{y}) = d_{\text{euc}}(\mathbf{y}, \mathbf{x})$; and a distance is never negative. They return in §7.

Euclidean distance is sensitive to overall magnitude. If a neuron population fires twice as hard to every image, every point moves twice as far from the origin and every distance doubles, even though the *pattern* across neurons is unchanged. The next two measures are designed to ignore that.

In NumPy: `np.linalg.norm(x - y)`.

Try it. What happens to $d_{\text{euc}}(\mathbf{x}, \mathbf{y})$ if both $\mathbf{x}$ and $\mathbf{y}$ are multiplied by 2?

5 Cosine similarity and cosine dissimilarity

Cosine similarity is $\cos \phi$ from §3.3: the dot product with both lengths divided out. It compares directions and ignores lengths. The course turns it into a dissimilarity (small means similar) by subtracting from one:

$$d_{\text{cos}}(\mathbf{x}^{(i)}, \mathbf{x}^{(k)}) = 1 - \cos \phi = 1 - \frac{\mathbf{x}^{(i)} \cdot \mathbf{x}^{(k)}}{\|\mathbf{x}^{(i)}\| \|\mathbf{x}^{(k)}\|}.$$

It is 0 when the two vectors point the same way, 1 when they are perpendicular, and 2 when they point in opposite directions; so it lies in $[0, 2]$. Both vectors must have nonzero length, or the fraction is undefined.

Example. $\mathbf{x}^{(1)} = (3, 4)$ and $\mathbf{x}^{(2)} = (4, 3)$: $\cos \phi = 0.96$, so $d_{\cos} = 0.04$. Nearly the same direction; nearly zero dissimilarity.

Example, where the two measures disagree. Take $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (3, 0)$. Euclidean distance: $\|(-2, 0)\| = 2$. Cosine: $\cos \phi = 3/(1 \cdot 3) = 1$, so $d_{\cos} = 0$. Euclidean says “two units apart”; cosine dissimilarity is zero because the directions match, although the vectors differ in length. Neither is wrong: one asks *how far apart are the points*, the other *do the arrows point the same way*. If $\mathbf{y}$ is the same neurons firing three times as hard, cosine ignores that gain and Euclidean distance reports it.

In NumPy: `1 - np.dot(x, y) / (np.linalg.norm(x) * np.linalg.norm(y))`, for one pair of vectors.

6 Correlation and correlation dissimilarity

6.1 Centring a vector by its own mean

The *mean* of a vector is the average of its entries, $\bar{x} = \frac{1}{D} \sum_j x_j$. *Centring* subtracts it from every entry:

$$\tilde{x}_j = x_j - \bar{x}, \quad \tilde{\mathbf{x}} = (x_1 - \bar{x}, \dots, x_D - \bar{x}),$$

so the centred vector $\tilde{\mathbf{x}}$ has mean zero. This is a per-vector operation: each stimulus is centred by its *own* mean, across its D entries.

Example. $\mathbf{x} = (1, 2, 3)$ has mean 2, so $\tilde{\mathbf{x}} = (-1, 0, 1)$. $\mathbf{y} = (2, 4, 6)$ has mean 4, so $\tilde{\mathbf{y}} = (-2, 0, 2)$. $\mathbf{z} = (3, 2, 1)$ has mean 2, so $\tilde{\mathbf{z}} = (1, 0, -1)$.

Example, a uniform offset disappears. $(1, 2, 4)$ has mean $7/3$ and $(6, 7, 9)$ has mean $22/3$; both centre to $(-\frac{4}{3}, -\frac{1}{3}, \frac{5}{3})$. Adding 5 to every entry changed the point but not the centred vector.

6.2 Pearson correlation is cosine similarity after centring

The *Pearson correlation* r between two vectors is the cosine of the angle between their centred versions:

$$r(\mathbf{x}, \mathbf{y}) = \frac{\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|} = \frac{\sum_j (x_j - \bar{x})(y_j - \bar{y})}{\sqrt{\sum_j (x_j - \bar{x})^2} \sqrt{\sum_j (y_j - \bar{y})^2}}.$$

The two formulas are equivalent: expand the centred vectors in the first to obtain the second. Both vectors must vary across their entries; a constant vector becomes $\mathbf{0}$ after centring and the fraction is undefined. r lies in $[-1, 1]$. It is 1 when $y_j = ax_j + b$ for every entry with $a > 0$, -1 when the same holds with $a < 0$, and 0 when there is no linear relation across these entries (a nonlinear relation may still exist).

Example. $\tilde{\mathbf{x}} = (-1, 0, 1)$ and $\tilde{\mathbf{y}} = (-2, 0, 2)$: $r = (2 + 0 + 2)/(\sqrt{2} \cdot \sqrt{8}) = 4/4 = 1$. Indeed $\mathbf{y} = 2\mathbf{x}$. $\tilde{\mathbf{x}} = (-1, 0, 1)$ and $\tilde{\mathbf{z}} = (1, 0, -1)$: $r = (-1 + 0 - 1)/(\sqrt{2} \cdot \sqrt{2}) = -1$. The pattern in $\mathbf{z}$ is the reverse of the pattern in $\mathbf{x}$.

The course's third dissimilarity is

$$d_{\text{corr}}(\mathbf{x}^{(i)}, \mathbf{x}^{(k)}) = 1 - r,$$

again in $[0, 2]$: 0 for a perfectly matching pattern, 1 for no linear relation, 2 for a perfectly reversed one.

Why centre? Multiplying either response vector by a positive constant leaves cosine similarity unchanged. Correlation is also unchanged when the same constant is added to every entry of either vector (a uniform offset within that vector, as in the example above). Correlation therefore compares only the *pattern* of ups and downs across neurons.

Example, where cosine and correlation disagree. Take $\mathbf{x} = (5, 5, 6)$ and $\mathbf{y} = (6, 5, 5)$. Both sit far from the origin in almost the same direction: $\mathbf{x} \cdot \mathbf{y} = 30 + 25 + 30 = 85$, $\|\mathbf{x}\| = \|\mathbf{y}\| = \sqrt{86}$, so $\cos \phi = 85/86 \approx 0.988$ and $d_{\text{cos}} \approx 0.012$. Now centre. Both means are $16/3$, so $\tilde{\mathbf{x}} = (-\frac{1}{3}, -\frac{1}{3}, \frac{2}{3})$ and $\tilde{\mathbf{y}} = (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})$. Then $\tilde{\mathbf{x}} \cdot \tilde{\mathbf{y}} = -\frac{2}{9} + \frac{1}{9} - \frac{2}{9} = -\frac{1}{3}$, each centred norm is $\sqrt{6/9}$, and

$$r = \frac{-1/3}{6/9} = -0.5, \quad d_{\text{corr}} = 1.5.$$

Cosine calls them nearly identical because both are dominated by a large shared baseline (every neuron firing at about 5). Correlation strips the baseline and sees what is left: the centred patterns are negatively correlated, because their largest entries occur at different neurons. This illustrates why positive response vectors, such as ReLU activations, can have high cosine similarity but much lower correlation.

In NumPy: `np.corrcoef(x, y)[0, 1]` gives r for one pair of vectors.

6.3 The three dissimilarities, side by side

For two response vectors $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(k)}$:

$$d_{\text{euc}} = \left\| \mathbf{x}^{(i)} - \mathbf{x}^{(k)} \right\|, \quad d_{\text{cos}} = 1 - \cos \phi, \quad d_{\text{corr}} = 1 - r.$$

measure	unchanged by	range	0 means	(3, 4) vs. (4, 3)
d_{euc}	nothing, in general	$[0, \infty)$	same point	$\sqrt{2} \approx 1.414$
d_{cos}	positive rescaling of either vector	$[0, 2]$	same direction	0.04
d_{corr}	positive rescaling plus a uniform offset	$[0, 2]$	$y_j = ax_j + b, a > 0$	2

The last cell is worth checking by hand: (3, 4) centres to $(-0.5, 0.5)$ and (4, 3) to $(0.5, -0.5)$, exactly opposite, so $r = -1$ and $d_{\text{corr}} = 2$. For two nonconstant vectors with two entries, correlation can only be $+1$ or -1 ; with three or more entries, intermediate values are possible.

7 The dissimilarity matrix

7.1 Definition

Choose one of the three measures and compute it for *every pair* of stimuli. The result is a square $N \times N$ matrix D , the *dissimilarity matrix* (the lectures also call it the RDM, representational dissimilarity matrix):

$$d_{ik} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)}), \quad i, k = 1, \dots, N.$$

Entry d_{ik} answers one question: *how dissimilar are stimuli i and k , under this measure, in this representation?* Row i of D is stimulus i 's dissimilarity to everything.

Example. Euclidean, on the three images of §4:

$$D = \begin{pmatrix} 0 & \sqrt{2} & \sqrt{18} \\ \sqrt{2} & 0 & \sqrt{20} \\ \sqrt{18} & \sqrt{20} & 0 \end{pmatrix} \approx \begin{pmatrix} 0 & 1.414 & 4.243 \\ 1.414 & 0 & 4.472 \\ 4.243 & 4.472 & 0 \end{pmatrix}.$$

7.2 Three checks for the RDMs used here

They follow from the three properties of a distance noted in §4, and they are the first thing to check when you have computed one:

- **Zero diagonal:** $d_{ii} = 0$. A stimulus is not dissimilar to itself.
- **Symmetric:** $d_{ik} = d_{ki}$, so $D = D^\top$ (the transpose, §9: the matrix with rows and columns swapped).
- **Non-negative:** $d_{ik} \geq 0$.

The same three hold for $d_{\cos}$ and d_{corr} , provided the measure is defined for every row: a zero-length row has no cosine dissimilarity and a constant row has no correlation. The Assignment 1 notebook guards the denominator with a tiny constant so the code does not divide by zero; that is a computational convention, not a definition of correlation for a constant vector, and the checks above assume it was not needed.

7.3 Counting the pairs

Because of the zero diagonal and the symmetry, an $N \times N$ dissimilarity matrix describes $N(N-1)/2$ unordered stimulus pairs, represented by the entries strictly above the diagonal (the *upper triangle*). For $N = 3$ that is 3 pairs; for $N = 120$ it is 7,140. When you compare two dissimilarity matrices, those are the entries you want: using all N^2 entries counts every pair twice and adds N zeros from the diagonal.

In NumPy: `np.allclose(D, D.T); np.allclose(np.diag(D), 0); (D >= 0).all()`.

8 Ranks and Spearman correlation

Sometimes only the *ordering* of a set of numbers is trustworthy, not their exact values. Rating scales are like this: the numerical difference between ratings 6 and 7 need not represent the same psychological difference as that between 2 and 3. The *rank* of an entry is its position when the entries are sorted from smallest to largest.

Example. $x = (10, 20, 25)$ has ranks $(1, 2, 3)$; $y = (1, 5, 4)$ has ranks $(1, 3, 2)$. Ties get the average of the positions they occupy: $(4, 4, 9)$ has ranks $(1.5, 1.5, 3)$.

The *Spearman correlation* ρ is the Pearson correlation of the two rank vectors. It measures whether two variables increase together, without assuming the relation is a straight line.

Example. Ranks $(1, 2, 3)$ and $(1, 3, 2)$ centre to $(-1, 0, 1)$ and $(-1, 1, 0)$, so $\rho = (1 + 0 + 0)/(\sqrt{2} \cdot \sqrt{2}) = 0.5$.

Two dissimilarity matrices are compared by taking the $N(N-1)/2$ upper-triangle entries of each, in the same pair order, and computing ρ between the two lists: a high ρ says the pairs one representation calls far apart, the other does too, whatever the units. Both matrices must refer to the same stimuli in the same row and column order; matching shapes alone does not establish this.

In NumPy: `scipy.stats.spearmanr(a, b).statistic`.

9 Optional: matrix products and the Gram matrix

9.1 What a matrix is, and two ways to read one

A matrix is a rectangular grid of numbers with m rows and n columns, “ $m \times n$ ”. Its entry in row i , column j is written with two subscripts, row first: $(A)_{ij}$ or a_{ij} . Two readings recur in this course. **As data:** X from §1.3, one row per stimulus, one column per feature. **As a map:** a matrix W transforms an input vector, $x \mapsto Wx$. This is the linear operation inside many neural-network layers (a layer usually adds a bias and applies a nonlinearity as well).

9.2 The matrix–vector product is a stack of dot products

If A is $m \times n$ and w has n entries, the product Aw is a vector with m entries, and its i -th entry is the dot product of row i of A with w :

$$(Aw)_i = \sum_{j=1}^n a_{ij} w_j = a_i \cdot w, \quad \text{where } a_i \text{ is row } i.$$

Example. With the 3×2 response matrix from §1.3 and $w = (1, -1)$, written as a column:

$$Xw = \begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3-4 \\ 4-3 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}.$$

Read it as “for each image, subtract neuron 2’s response from neuron 1’s response”: one number per image. The inner dimensions must agree: a 3×2 matrix takes a vector with 2

entries and returns one with 3. A matrix times a matrix works the same way, one column at a time: $(m \times k)(k \times n) \rightarrow (m \times n)$.

In NumPy: `A @ w`; the shapes must read $(m, n) @ (n,) \rightarrow (m,)$.

9.3 The transpose, and every dot product at once

The *transpose* $A^\top$ swaps rows and columns: $(A^\top)_{ij} = a_{ji}$, and an $m \times n$ matrix becomes $n \times m$.

$$\mathbf{X} = \begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 0 & 1 \end{pmatrix} \implies \mathbf{X}^\top = \begin{pmatrix} 3 & 4 & 0 \\ 4 & 3 & 1 \end{pmatrix}.$$

Transposing a data matrix turns “one row per stimulus” into “one column per stimulus”. One product is worth knowing by name: $\mathbf{X}\mathbf{X}^\top$ (the *Gram matrix*) is $N \times N$ and its entry (i, k) is the dot product of row i with row k , that is, $\mathbf{x}^{(i)} \cdot \mathbf{x}^{(k)}$, every pairwise dot product at once.

$$\mathbf{X}\mathbf{X}^\top = \begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 & 0 \\ 4 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 25 & 24 & 4 \\ 24 & 25 & 3 \\ 4 & 3 & 1 \end{pmatrix}.$$

The diagonal holds the squared norms (25, 25, 1, matching §3.2) and the off-diagonal entries are the dot products computed in §3.1. A matrix that equals its own transpose, as this one does, is called *symmetric*.

In NumPy: `X.T`; `X @ X.T`.

Answers to “Try it”

§1.3: row 3, column 2, which is $x_2^{(3)} = 1$. §4: it doubles, because $\|2\mathbf{x} - 2\mathbf{y}\| = 2\|\mathbf{x} - \mathbf{y}\|$; $d_{\cos}$ and d_{corr} are unchanged.

Symbols used in this handout

$\mathbf{x}, x_j$	a vector; its j -th entry	$\ \mathbf{x}\ $	Euclidean norm, $\sqrt{\mathbf{x} \cdot \mathbf{x}}$
D, N	entries per vector; number of stimuli	$\phi, \cos \phi$	angle between vectors; cosine similarity
$\mathbf{x}^{(i)}, x_j^{(i)}$	response to stimulus i ; entry j	$\bar{\mathbf{x}}, \tilde{\mathbf{x}}$	mean of $\mathbf{x}$; $\mathbf{x}$ centred by its own mean
$\mathbf{X}$	$N \times D$ response matrix	r, ρ	Pearson; Spearman (Pearson on ranks)
$\mathbf{x} \cdot \mathbf{y}$	dot product, $\sum_j x_j y_j$	$d_{\text{euc}}, d_{\cos}, d_{\text{corr}}$	$\ \mathbf{x} - \mathbf{y}\ , 1 - \cos \phi, 1 - r$
$A^\top$	transpose, rows and columns swapped	D, d_{ik}	$N \times N$ dissimilarity matrix; entry (i, k)